

PRST (08/09/2021)

$$X \text{ rv} \rightarrow M_X(t) = E(e^{tX})$$

$$a+bX \quad E(e^{t(a+bX)}) = E\left(e^{at} \cdot e^{btX}\right)$$

$$= e^{at} E(e^{btX})$$

$$= e^{at} M_X(bt)$$

If X_1, X_2 are rv which are independent $M_{X_1+X_2}(t) = M_{X_1}(t) M_{X_2}(t)$.

Characteristic function (chf)

$$\phi_X(t) = E(e^{itX})$$

$i = \sqrt{-1}$
 t - any real value.

If Mgf exists, $\phi_X(t) = M_X(it)$.
for a distribution F , chf is unique and vice versa.

$$\phi_X(0) = 1.$$

Ex 1. X is a discrete rv with pmf $f(x) = \frac{e^{-\lambda} \lambda^x}{x!}, x=0,1,2,\dots$

$$\phi_X(t) = ? \quad E(e^{itX})$$

$$\phi_X(t) = e^{(\lambda e^{it} - \lambda)}$$

Ex 2.

$$X = \begin{cases} -1 & \text{wp } \frac{1}{2} \\ +1 & \text{wp } \frac{1}{2} \end{cases}$$

with probability

$$P(X=-1) = \frac{1}{2}$$

$$P(X=1) = \frac{1}{2}$$

$$\text{chf } \phi_X(t) = ?$$

$$\phi_X(t) = \frac{1}{2} e^{-it} + \frac{1}{2} e^{it}$$

$$= \frac{1}{2} (e^{it} + e^{-it})$$

$$= \cos t.$$

$$e^{it} = \cos t + i \sin t$$

Joint Distribution

we toss the coin 4 times.

$$\omega_1 = HHHH,$$

$$\omega_2 = HHHT$$

$\vdots$

$$\omega_{16} = TTTT$$

X - a rv denoting the number of heads in any outcome.

looking the longest length of H's.

Y - arr denoting the longest sum of T's.

joint probability distribution for X, Y .

$X \backslash Y$	0	1	2	3	4	$P(X=x)$
0	$\frac{1}{16}$	0	0	0	0	$\frac{1}{16}$
1	0	$\frac{4}{16}$	0	0	0	$\frac{4}{16}$
2	0	$\frac{3}{16}$	$\frac{3}{16}$	0	0	$\frac{6}{16}$
3	0	0	$\frac{2}{16}$	$\frac{2}{16}$	0	$\frac{4}{16}$
4	0	0	0	0	$\frac{1}{16}$	$\frac{1}{16}$
Total	$\frac{1}{16}$	$\frac{7}{16}$	$\frac{5}{16}$	$\frac{2}{16}$	$\frac{1}{16}$	1

$P(X=0, Y=0)$
 $P(W = TTTT)$
 $P(X=0, Y=1)$
 $P(X=1, Y=1)$
 $P(X=2, Y=1)$
 $P(X=3, Y=1)$
 $P(X=4, Y=1)$

$P(Y=0)$ $P(Y=1)$ $P(Y=4)$

HTTT
 THTT
 TTHT
 TTTT

Y	$P(Y=y)$
0	$\frac{1}{16}$
1	$\frac{7}{16}$
2	$\frac{5}{16}$
3	$\frac{2}{16}$
4	$\frac{1}{16}$

→ Marginal distribution of Y .

Conditional distribution

$$P(X=0 / Y=4) = \frac{P(X=0, Y=4)}{P(Y=4)} = 0$$

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X	$P(X=x / Y=4)$
0	0
1	0
2	
3	
4	

conditional dist of X given $Y=4$

If X and Y are independent

$$P(X=x / Y=y) = P(X=x)$$

$$\Rightarrow P(X=x, Y=y) = P(X=x)P(Y=y)$$

$\forall x, y$

$$E(X, Y) = \sum_i \sum_j x_i y_j P_{ij}$$

$$P_{ij} = P(X=x_i, Y=y_j)$$

$Y \backslash X$	5	7	total
3	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{3}$
6	$\frac{5}{12}$	$\frac{1}{4}$	$\frac{2}{3}$
Total	$\frac{1}{2}$	$\frac{1}{2}$	1

$$E(XY)$$

$$E(XY) = 5 \times 3 \times P(X=5, Y=3) + 5 \times 6 \times P(X=5, Y=6) + 7 \times 3 \times P(X=7, Y=3) + 7 \times 6 \times P(X=7, Y=6)$$

$$= 29.5$$

$$E(g(X, Y)) = \sum_i \sum_j g(x_i, y_j) P_{ij}$$

$$g(X, Y) = \left(\frac{X^2}{Y} \right)$$

$$\sin\left(\frac{X}{Y}\right)$$

$$\text{Cov}(X, Y) = E\left[(X - \mu_X)(Y - \mu_Y)\right] \quad g: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\mu_X = E(X) \quad \mu_Y = E(Y) \quad = E(XY) - \mu_X \mu_Y$$

?

Ex

$Y \backslash X$	5	7	total
3	a	$\frac{1}{3} - a$	$\frac{1}{3}$
6	$\frac{1}{2} - a$	$a + \frac{1}{6}$	$\frac{2}{3}$
total	$\frac{1}{2}$	$\frac{1}{2}$	

$$\text{Cov}(X, Y) = -\frac{1}{2}$$

$$? \quad a = ?$$

what are the joint probabilities.